

In studies of Guillain-Barré syndrome, functional deficit is usually assessed according to a functional scale consisting of several categories. The level of interobserver agreement in this scoring method is not known; furthermore, this method seems to be insensitive when applied to bedridden and artificially ventilated patients. We have developed an additional score (MRC-sumscore), reflecting muscle strength in general. Both scoring methods, tested in Guillain-Barré patients, have an almost perfect interobserver agreement. For the functional score $\kappa = 0.85$, and for the MRC-sumscore $r^2 = 0.96$. The MRC-sumscore is easily assessed and more sensitive than the functional score when patients are bedridden or artificially ventilated.

Key words: Guillain-Barré syndrome • muscle strength • functional abilities

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INTEROBSERVER AGREEMENT IN THE ASSESSMENT OF MUSCLE STRENGTH AND FUNCTIONAL ABILITIES IN GUILLAIN-BARRÉ SYNDROME

RUUD P. KLEYWEG, MD, PhD, FRANS G. A. VAN DER MECHÉ, MD, PhD, and PAUL I. M. SCHMITZ, PhD

In several studies on the effect of plasma exchange in Guillain-Barré syndrome (GBS), functional deficit is assessed using a categorized scale.^{2,3} The scale, used in the North American trial,³ was originally developed by Hughes,⁵ and later slightly modified by others.^{3,6} It has the advantage that each change in category is of definite clinical importance. Moreover, most steps between the categories are well defined. Therefore, one might well expect a high interobserver agreement, something which is particularly important in multicenter studies. This scale is, however, rather insensitive to early clinical changes, as has recently been demonstrated in GBS patients in whom secondary clinical deterioration, after an initially suc-

cessful response to plasma exchange, was not always indicated by a change in functional score.⁹

An ideal grading system should be easily applicable and relevant to the clinical condition; there should be a high intra- and interobserver agreement, and the sensitivity should be sufficiently high to detect relevant changes.

In the Dutch Guillain-Barré trial, comparing high-dose immunoglobulins with plasma exchange, we introduced a second score, which had been tested in a pilot study.⁶ This score is a summation of the strength of 6 muscle groups tested on both sides (Table 1) according to the Medical Research Council (MRC) scale (Table 2); it yields the so-called "MRC-sumscore," ranging from 0 (paralysis) to 60 (normal strength). These muscle groups can easily be tested against gravity, and both proximal and distal muscle groups are repre-

From the Department of Neurology, University Hospital Rotterdam-Dijkzigt (Drs. Kleyweg and van der Meché), and Department of Biostatistics, Erasmus University, Rotterdam, The Netherlands (Dr. Schmitz).

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Address reprint requests to R. P. Kleyweg, MD, PhD, Department of Neurology, Merwede Ziekenhuis-Dordrecht, Postbus 306, 3300 AH Dordrecht, The Netherlands.

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Table 1. Muscle groups (right and left) assessed in the measurement of the MRC-sumscore.

Abduction of the arm
Flexion of the forearm
Extension of the wrist
Flexion of the leg
Extension of the knee
Dorsal flexion of the foot

Table 2. MRC-scale with full figures only. The patient is investigated in sitting posture and/or lying supine.

0	= No visible contraction
1	= Visible contraction without movement of the limb (not existent for hip flexion)
2	= Movement of the limb but not against gravity
3	= Movement against gravity over (almost) the full range
4	= Movement against gravity and resistance
5	= Normal

sented in the MRC-sumscore. The MRC-sumscore ranges from 0 (paralytic) to 60 (normal strength).

In the present study, we determined the sensitivity and interobserver agreement of both scoring systems. Furthermore, we studied the relationship between the MRC-sumscore and the grip strength measured on a continuous scale by dynamometry.

PATIENTS AND METHODS

All GBS patients studied in the Dutch Guillain-Barré trial were examined according to a pre-defined schedule for a period of 6 months. Initially, examination took place 3 times a week, the frequency gradually decreasing to once a month at the end of the follow-up period. Functional score and MRC-sumscore were assessed at each examination. The functional score was assessed using a grading system from 0 to 6: 0 = healthy; 1 = minor symptoms and signs, fully capable of manual work; 2 = able to walk >10 m without any assistance; 3 = able to walk >10 m with a walker or support; 4 = bed or chair-bound (unable to walk >10 m with a walker or support); 5 = assisted ventilation required for at least part of the day; and 6 = dead.^{3,5,6}

For the interobserver analysis, 2 groups of GBS patients were studied. The first group, the "trial group," consisted of 60 patients of the Dutch Guillain-Barré trial, consecutively tested by one of us (RK). In this multicenter trial, high-dose immunoglobulins (Gammagard, Baxter) were compared with plasma exchange in patients with GBS who became unable to walk independently within 2 weeks of onset. In order to control whether there was any bias in this nonblinded trial, one of the coordinators examined every patient in the co-operating centers. At that time, he was unaware of the treatment modality. The functional score and the MRC-sumscore were assessed by the coordinator and treating neurologist on the same day, independently of each other. After documentation, the results were discussed as part of a continuous program to optimize the scoring methods.

The second group, the "resident group," consisted of 15 consecutive GBS patients all admitted to the Academic Hospital of Rotterdam between November 1987 and January 1989.

Ten senior residents of the Neurology Department, all of whom were familiar with testing muscle strength according to the MRC-scale, were combined to form 30 different pairs. Each pair examined 1 patient twice with an interval of several weeks. Before each examination, the patient was seen by one of us (RK) who assessed the functional and MRC-sumscore and explained the procedure to the patient. The patient was examined independently by both members of the pair on the same day, but they were unaware of each other's findings. During each examination, the functional score and MRC-sumscore were assessed and written on a standardized data form. We explained thoroughly how the patients should be scored at the beginning of the study. In total, 60 paired observations were made of the patients at different stages of their disease.

In order to examine the relationship between the MRC-sumscore and the grip strength, 11 GBS patients admitted to the Academic Hospital of Rotterdam were followed longitudinally by one of us (RK) for several months, measuring MRC-sumscore and grip strength at the same time.

STATISTICAL METHODS

Calculation of interobserver agreement was performed for the "trial group" as well as for the "resident group." For the functional score, we used (κ) statistics, which correct the level of observed agreement for chance agreement.¹ Intra-class correlation coefficients were calculated for the MRC-sumscore.⁷ Furthermore, we assumed that a difference in MRC-sumscore of less than 10% of the total range (i.e., a difference of less than 6 points) between two observers would be acceptable and, therefore, judged as agreement. A difference of more than 10% implies disagreement.

In order to compare the MRC-sumscore and grip strength measured by hand dynamometry, we calculated Fisher's correlation coefficient.

RESULTS

Sensitivity of the Functional Score and MRC-Sum-score. The sensitivity of the functional score and MRC-sumscore was studied by measuring these scores simultaneously at different stages of the disease. In Figure 1, the clinical course for a single

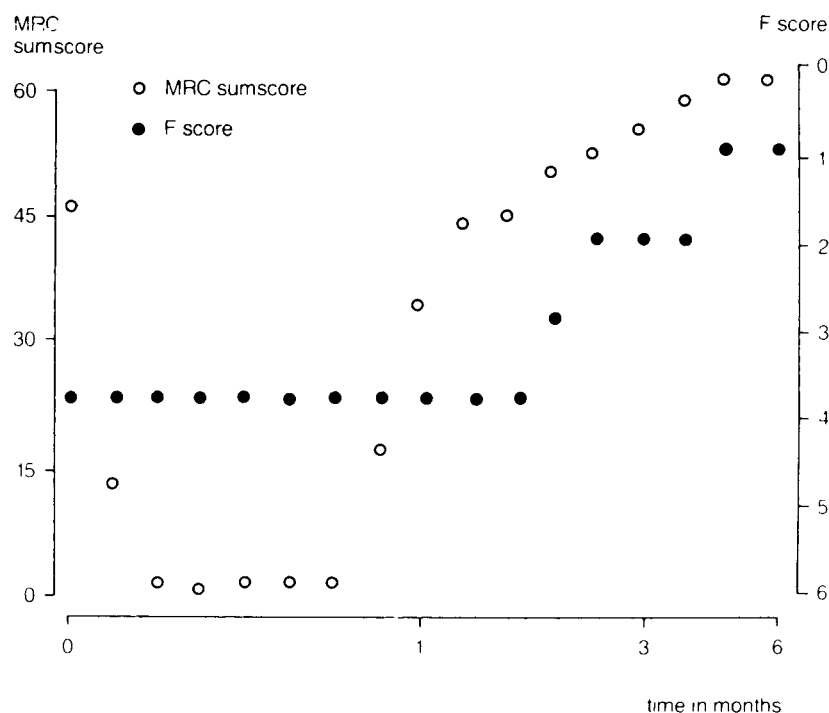


FIGURE 1. Clinical course as depicted by MRC-sumscore and functional (*F*) score.

patient is shown as reflected in MRC-sumscore and functional score. When the patient is still deteriorating, the MRC-sumscore declines, while the functional score remains the same. Recovery is also noticed first in the MRC-sumscore and only later in the functional score. After the first 100 patients had participated in the Dutch GBS trial, it was assessed that about 30% of the patients showed such a difference in sensitivity between the course of the functional score and the MRC-sumscore.

We also analyzed the range of MRC-sumscores of all patients in a certain functional group. Figure 2 shows the relationship between the functional score and the MRC-sumscore of 60 patients in the “trial group” measured by both observers. Within functional groups 1, 2, and 3, there is limited variation in the MRC-sumscore, whereas in functional groups 4 and 5, the MRC-sumscore ranges from 0 to 55. The same pattern is seen in the observations in the “resident group.” This pattern does not change if the functional score is related to the MRC-sumscore of the arms or legs separately.

Interobserver Agreement on the Functional Score. In the “trial group” there was agreement on the functional score in 53 of 60 paired scores (88.3%)

(Fig. 3a); correction for chance results in $\kappa = 0.85$.

Figure 3b shows the results of the study in the “resident group.” Here, the observed agreement is 86.7%, and after correction for chance, $\kappa = 0.80$. The maximum difference in functional score was 1 point, occurring predominantly between functional scores 1 and 2.

Interobserver Agreement on the MRC-Sumscore.

Figure 4a shows the correlation between the MRC-sumscores measured in the “trial group.” The agreement is expressed in an intraclass correlation coefficient which is 0.98 ($r^2 = 0.96$). The relationship between the MRC-sumscores measured in the patients examined by different pairs of residents is reflected in Figure 4b; the intraclass correlation coefficient is 0.96 ($r^2 = 0.93$).

If we assume that a difference of less than 10% (less than 6 points on the scale 0–60) between the 2 MRC-sumscores means agreement, then there is agreement between 2 observers in the “trial group” on 90% of the observations, and on 78% between the residents. In this latter group, there was 1 observer who scored systematically lower than the others. If one disregards this observer’s results, then there is good agreement on 89% of the observations.

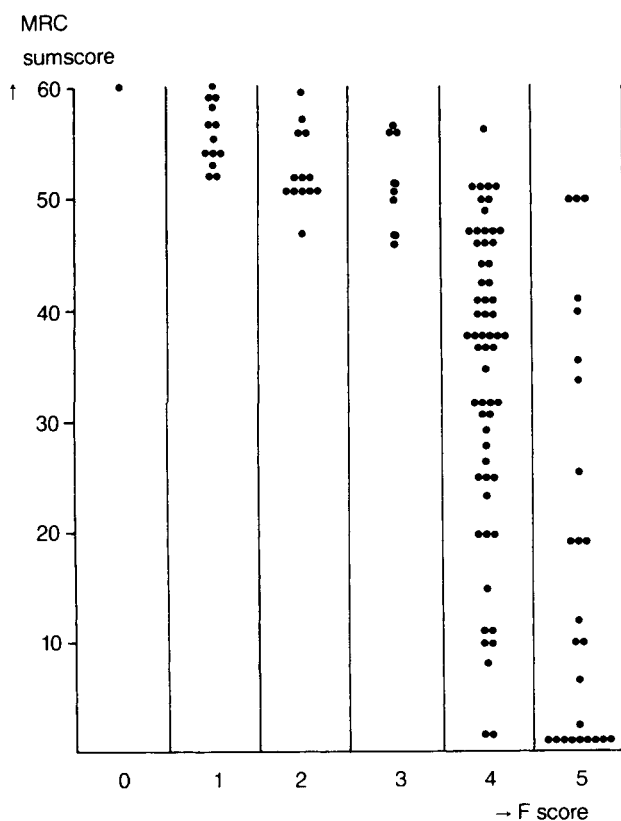


FIGURE 2. Relationship between functional (*F*) score and MRC sumscore in 60 patients, measured by 2 observers at the same arbitrarily chosen time in the course of the disease.

The Relationship Between Grip Strength and MRC-Sumscore. Eleven patients were followed over a period of several months, with regular measurements of the grip strength and MRC-sumscore during the same examination. The course of the MRC-sumscore and grip strength in the individual patient often ran parallel (Fig 5), and the correlation coefficients between grip strength and MRC-sumscore for these patients ranged from 0.80 to 0.99 with a median of 0.91.

DISCUSSION

A considerable change in the clinical condition can occur, as reflected by the MRC-sumscore, without any accompanying change in functional score. Even if chair-bound or bedridden on admission, a patient can still be quite strong. He or she may subsequently become tetraparalytic and then improve again without any change in functional score (Fig. 1). In general, it appears that functional grades 4 and 5 are compatible with a wide range of MRC-sumscores (Fig. 2). In contrast, in functional scores 1–3, the MRC-sumscores are in a narrower range. One may, therefore, conclude that, especially for patients who are bedridden or receiving artificial ventilation (grades 4 and 5), measurement of the MRC-sumscore contributes valuable information about the clinical course. In the higher functional categories,^{1–3} the MRC-sumscore adds little extra information.

There are several possible explanations for the

Functional category

↓ →	5	4	3	2	1	0
5	11	-	-	-	-	-
4	-	14	-	-	-	-
3	-	1	10	-	-	-
2	-	-	2	9	-	-
1	-	-	-	4	9	-
0	-	-	-	-	-	-

A Kappa = 0.85

Functional category

↓ →	5	4	3	2	1	0
5	12	-	-	-	-	-
4	1	29	-	-	-	-
3	-	1	4	-	-	-
2	-	-	1	3	-	-
1	-	-	-	4	4	-
0	-	-	-	-	1	-

Kappa = 0.80

B

FIGURE 3. Observed agreement in functional score between observers in the “trial group” (A), and between observers in the “resident group” (B).

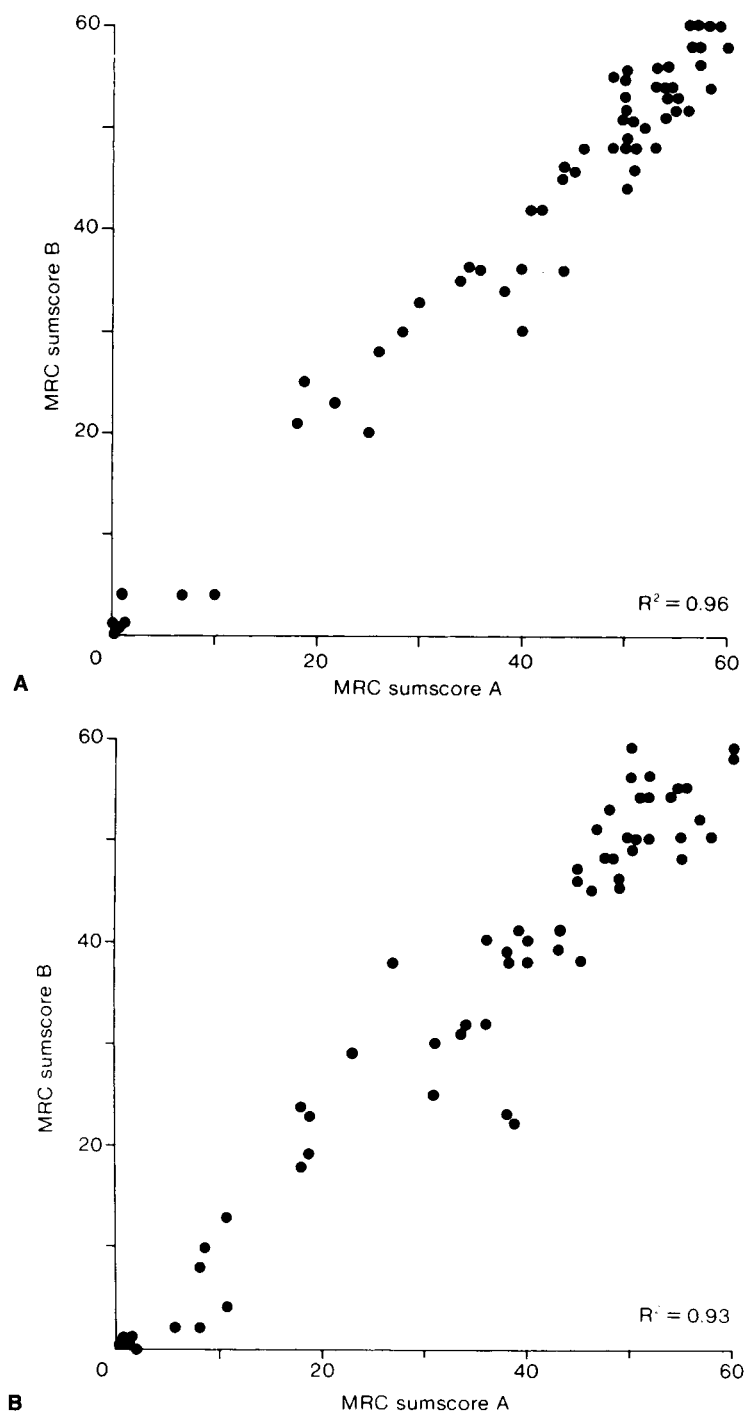


FIGURE 4. Relationship between MRC-sumscore measured by 2 observers in the “trial group” (A), and measured by two observers in the “resident group” (B).

wide range of MRC-sumscores in functional grades 4 and 5. As the functional score is mainly an indication of the neurological condition of the legs, the change in MRC-sumscore without a change in functional score might be explained by a progressive decline in strength of the arm mus-

cles, whereas the muscles of the legs were already weak at onset. This contention has, however, not been supported by analysis of the MRC-sumscores of arms and legs separately.

Other possible explanations for being bed- or chair-bound (functional score 4), despite consider-

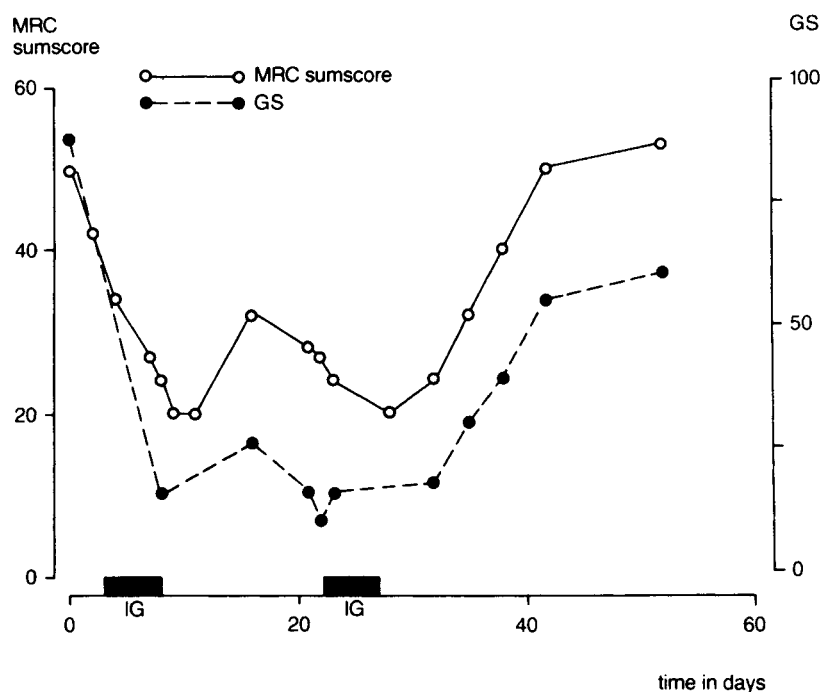


FIGURE 5. Relationship between the MRC-sumscore and grip strength (GS) in a GBS patient who showed secondary deterioration after an initial response to high-dose immunoglobulin therapy (IG). After a second treatment course he improved again.

able strength, are the presence of severe sensory deficits or predominantly proximal weakness. Furthermore, in the ventilated group, respiratory muscles may be more selectively involved. Also, complications of artificial ventilation may extend the time spent on the respirator, despite improved muscle strength.

To be useful in further studies, interobserver agreement on these scoring methods must be high. Interobserver agreement can be expressed in a κ value which corrects the level of observed agreement for chance agreement; κ values of 0.80 and higher indicate perfect agreement according to the guidelines of Landis and Koch.⁸ For the functional score, there was almost perfect agreement in both study groups, the value for the trial group being slightly better. The distinction between functional categories 1 and 2 is sometimes difficult, as was also reported in a recent trial.³

Measurement of the MRC-sumscore resulted in high intraclass correlation coefficients of 0.98 and 0.96, indicating excellent agreement. If one considers a difference of less than 10% in MRC-sumscore between 2 observers to be good agreement, then the "trial group" has agreement in 90% of the observations versus 78% in the resident group.

In the paired observations with a considerable difference in MRC-sumscore, the patients were often being artificially ventilated and some were receiving sedatives, leading to fluctuation of cooperation during the day.

The difference in agreement between the 2 groups might be explained by the fact that in the trial group one observer was always the same (RK); moreover the investigator always received feedback in order to optimize the scoring within the trial. In the other group, the residents were informed only at the beginning of the study and were not usually familiar with the patient. Moreover, in this group there was one observer who consistently scored considerably lower than the scoring partner. Such a systematic difference in scores between 2 observers is not necessarily bad. If progression or recovery has to be documented, one observer may, for instance, score a change in MRC-sumscore from 10 to 20, and another observer may score the same patient initially with MRC-sumscore 16, subsequently improving to MRC-sumscore 26. In this case, there is excellent agreement about the degree of improvement.

One might argue that the MRC-sumscore is a very crude means of assessing muscle strength. Each muscle group is scored on a very nonlinear,

ordinal scale; grade 4, in particular, spans a large range of muscle strength. One cannot, therefore, deny that dynamometers or myometers would give a more exact assessment of strength. These methods are, however, much less practical for daily bedside examination and require extensive standardization and training before usage. Furthermore, this method is not suitable for very weak muscles (MRC grade 1–3).

In general practice, we and many other neurologists use a more differentiated MRC-score in the 3 to 5 range using +, −, and ½ to overcome the wide variation in condition described by MRC-score 4. Interobserver agreement on this extended MRC-score was not been determined but is probably lower than that for the original MRC-score. This differentiated score is important in the follow-up of individual muscle groups. However, for our purpose, (i.e., follow-up of the general condition of patients in a multicenter trial), these refinements proved not to be necessary.

When we compare the MRC-sumscore with a hand dynamometer score, there is generally a strong correlation between these two scores during the acute, plateau, and recovery phases. The MRC-sumscore has proved to be a reliable way of obtaining an overall impression of muscle strength and one might, according to the central limit theorem,⁴ regard the MRC-sumscore as a continuous variable.

In conclusion, both the functional score and the MRC-sumscore have an almost perfect interobserver agreement; this might even be improved by regular feedback about the results. The MRC-sumscore is more sensitive to changes in clinical course than the functional score in the more severe functional grades. If one uses this MRC-sumscore in addition to the functional score, it is possible to follow the clinical course in more detail and recognize when recovery begins and a secondary relapse occurs.

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